

PARTS LIST

Semiconductors:

- IC1 - TL072 dual JFET
op-amp/amplifier
IC2 - TDA8561Q radio power
amplifier
LED1 - 5mm LED
T1, T2 - BC547 npn transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R4 - 1-k Ω -ohm
R2, R3, R5, R6, R19 - 10-k Ω -ohm
R7, R8 - 100-k Ω -ohm
R9, R10, R13-R15, R17 - 47-k Ω -ohm
R11, R12 - 15-k Ω -ohm
R16 - 6.8-k Ω -ohm
R18 - 1.5-k Ω -ohm
VR1 - 10-k Ω -ohm log potmeter
VR2 - 10-k Ω -ohm dual-gang
potmeter

Capacitors:

- C1-C3, C6 - 4.7 μ F, 25V electrolytic
C4, C5 - 68nF ceramic disk
C7 - 33nF ceramic disk
C8, C9 - 220 μ F, 25V electrolytic
C10 - 220nF ceramic disk
C11 - 100nF ceramic disk
C12 - 2200 μ F, 35V electrolytic

Miscellaneous:

- CON1 - 2-pin terminal block
LS1 - 8-ohm, 24W woofer
RCA1, RCA2 - RCA socket
SI A, SI B - SPDT switch
S2 - On/off switch

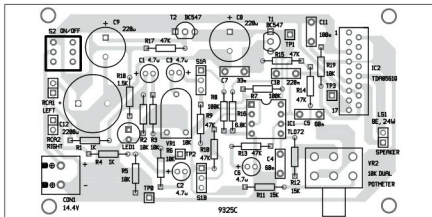


Fig. 3: Component layout for the PCB

Construction and testing

An actual-size, single-side PCB for the subwoofer is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on PCB, enclose it in a suitable plastic case. Use suitable heat sink for IC2.

Fix 2-pin connector for connecting 8-ohm, 24W woofer LS1 and connect the power supply at CON1. The left

and right stereo channels are to be connected to RCA1 and RCA2 through external wires. Ensure that VR1 and VR2 are accessible from outside for volume and LPF frequency control. Use voltage levels mentioned in the test points table for debugging, if required. ●

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